IN A JACOBSON ALGEBRAIC SPACE, POINTS OF FINITE TYPE LIFT TO CLOSED POINTS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Scheme case. Morphisms, lemma Jacobson-points-lift (cross-chapter, named). The scheme case for the etale cover.
- Points of the presentation. Properties of Spaces, lemma points-cartesian (cross-chapter, named). The fibre identification.

Source reference. Original lemma 0BA3: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5467–5467. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02J4: focused-seeds/geometry/S9-decent-jacobson-ft-points-lift-to-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TB: focused-seeds/geometry/S9-decent-jacobson-ft-points-lift-to-closed/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06EH: focused-seeds/geometry/S9-decent-jacobson-ft-points-lift-to-closed/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a Jacobson algebraic space over S. For $x \in X_{ft\text{-}pts}$ and $g : W \to X$ locally of finite type with W a scheme, if $x \in \mathrm{Im}(|g|)$, then there exists a closed point of W mapping to x.*

Proof. Let $U \to X$ be an étale morphism with U a scheme and with $u \in U$ closed mapping to x, see Morphisms of Spaces, Lemma 06EH. Observe that W, $W \times_X U$, and U are Jacobson schemes by Lemma 0BA3. Hence finite type points on these schemes are the same thing as closed points by Morphisms, Lemma 01TB. The inverse image $T \subset W \times_X U$ of u is a nonempty (as x in the image of $W \to X$)

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closed subset. By Morphisms, Lemma 02J4 there is a closed point t of $W \times_X U$ which maps to u. As $W \times_X U \to W$ is locally of finite type the image of t in W is closed by Morphisms, Lemma 01TB. $\square$